Case 1 — Mr. Samuel Okoro

68 M | 5 days post-ischaemic stroke — Right MCA | Acute Stroke Unit, medically stable | Left hemiparesis (UL + face) Left-sided neglect Pusher syndrome Poor trunk control 2-person assist STS Day 5 acute

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Left lower facial weakness (UMN pattern)
- Trunk control deficit — unable to sit unsupported
- Presumed left sensory impairment (proprioception, light touch)
- Unable to bear weight through left LL safely
- Reduced insight / anosognosia:

■■ RED FLAGS

- New or worsening neurological deficit
- Dysphagia / wet voice / coughing on fluids:
- Immediate aspiration risk — refer urgently to SLT; do not allow oral intake without SLT clearance.
- Cardiovascular instability
- (BP >180/110 mmHg or <90/60 mmHg, resting HR >100, SpO■ <94%): Physiotherapy is contraindicated — notify the nurse in charge and await medical review before any active rehabilitation.
- Unilateral calf swelling, warmth, or redness in the hemiplegic leg:
- Suspected DVT — mobilisation risks pulmonary embolism; urgent Doppler ultrasound and medical review required.
- Severely altered consciousness or acute agitation:

■ PATHOPHYSIOLOGY (KEY POINTS)

- Occlusion of the right MCA results in infarction of the corticospinal tract fibres subserving the contralateral (left) body, producing upper motor neuron hemiparesis.
- Sign / Symptom Pathophysiological Explanation Left-sided hemiparesis (UL > LL) Right motor cortex / corticospinal tract infarction.
- UL fibres occupy a more lateral and superior position in MCA territory, hence disproportionately affected by MCA occlusion compared with the more medially represented LL fibres (ACA territory).
- Left lower facial weakness UMN lesion — contralateral lower face only, because the forehead (frontalis) has bilateral cortical representation.
- Classic distinction from LMN facial palsy which affects all ipsilateral facial muscles.
- Left hemispatial neglect Right hemisphere parietal / fronto-parietal network damage.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations Before Testing

Tone:

Passive movement of UL and LL — expect flaccid / hypotonic acutely. Use Modified Ashworth Scale. Note any early spasticity emerging in wrist flexors or ankle plantar flexors.

Posture: In supported sitting — observe direction and magnitude of trunk lean, active pushing against the plinth toward the left, and head position. Does he perceive himself as upright? Neglect: Observe spontaneous attention — does he orient only to the right? Does he fail to notice or reach for items on the left? Does he leave the left arm in unsafe positions? Coordination:

Test 1 — Scale for Contraversive Pushing (SCP)

What it measures: The severity of pusher syndrome — specifically, the degree to which the patient spontaneously pushes toward the hemiparetic side and actively resists passive correction back to midline. Why I choose it:

■ TREATMENT PLAN

- "Do you have any pain — particularly in your left shoulder or arm?"
- "Have you had any difficulty swallowing food or drinks, or any changes to your speech?"
- "Before all of this happened, how were you getting around — were you independent at home?"
- Observe the patient in sitting (and standing if tolerated) without manual correction from the therapist.
- Posture subscale (0–1):
- Is there spontaneous lateral lean toward the affected side? Score 0 = no lean; 0.25 = slight lean; 0.75 = moderate; 1 = extreme lean.
- Extension subscale (0–2):
- Does the patient use the non-paretic arm or leg to actively push (extend and abduct)? Score 0 = no; 1 = in one position; 2 = in both sitting and standing.
- Resistance subscale (0–1):
- When the therapist attempts to passively correct the patient toward true midline, does the patient resist? Score 0 = no resistance; 1 = resistance present.
- Sum all subscale scores. Total range 0–4; score ≥1 = pusher syndrome present.
- Positive finding: